



VICTORIA JUNIOR COLLEGE
BIOLOGY DEPARTMENT
JC2 PRELIMINARY EXAMINATIONS 2013
Higher 2

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CANDIDATE NAME

CLASS

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INDEX NUMBER

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BIOLOGY

9648/03

Applications Paper and Planning Question

Paper 3

20 September 2013

Additional Materials: Answer Booklet/ Paper

2 Hours

READ THESE INSTRUCTIONS FIRST

Write your CT GP/ INDEX NO. and name on all the work you hand in.

Write in dark blue or blue pen.

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use any staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

At the end of the examination, fasten all your work securely together.

The intended number of marks is given in brackets [] at the end of each question.

For Examiner's Use	
Section A	
1	
2	
3	
4	
5	
Total	

- 1 A violent air crash resulted in the death of 200 passengers. The poor condition of the crash victims made physical identification impossible. Forensic scientists were tasked to confirm the identities of victims by using molecular techniques for families who have come forward to claim the correct remains of their relatives.

DNA samples of human remains and surviving relatives were subjected to restriction enzyme digest and Southern blot analysis to analyse VNTR band patterns. Single locus VNTR probes will yield DNA patterns of seven individuals at a single locus (**Fig 1.1**). On the other hand, multi-locus VNTR probes can yield more information about each individual since 10 to 30 loci can be simultaneously analysed. This is illustrated in five individuals in **Fig.1.2**.

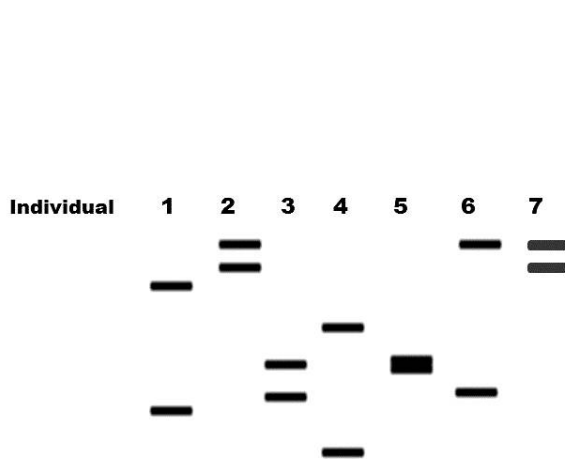


Fig 1.1

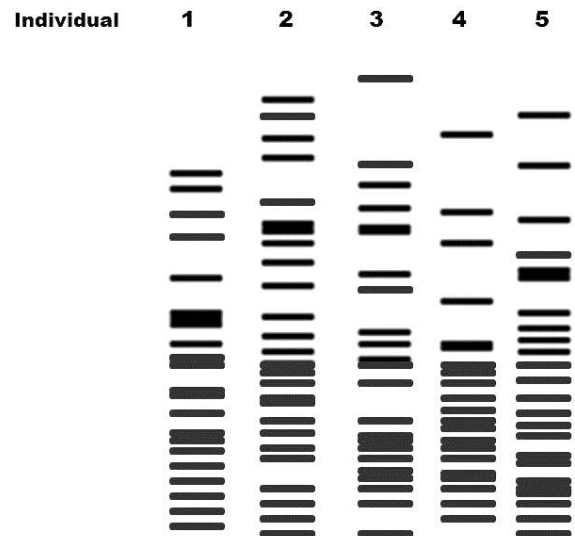


Fig 1.2

- (a) (i) Explain the basis of application of VNTR in DNA profiling.

[3]

- (ii) With reference to **Fig. 1.1**, explain the different numbers of fragments seen in different individuals.

[2]

(iii) With reference to **Fig. 1.2**, suggest why multi-locus probes are used.

[2]

(b) VNTRs can be analysed by Southern blotting or by PCR analysis. Describe **two** differences between these techniques in the analysis of the single locus VNTR shown in **Fig. 1.1**.

[4]

(c) Other than DNA profiling, RFLP analysis was used in the Human Genome Project (HGP). Outline how information from its application helped in obtaining the map of the human genome.

[2]

- (d) The fully sequenced human genome contains many unknown genes. Suggest how RFLP analysis can help to identify **unknown** genes that are responsible for heritable diseases.

[2]

[Total: 15]

- 2** Glioblastoma multiforme is the most aggressive form of human brain tumour, which has no effective cure.

In an attempt to find a treatment for glioblastoma, researchers used adenoviruses carrying the partial gene sequence of telomerase to inhibit the growth of the tumour. This sequence codes for the 251 amino acids from the C-terminal end of the human telomerase reverse transcriptase (hTERTC27). Previous experiments on cervical cancer cells show that when this sequence is overexpressed, a dysfunctional telomerase result and this can cause the telomeres of the cancerous cells to rapidly decrease over the course of cell divisions, which can induce senescence.

The hTERTC27 is produced using PCR (polymerase chain reaction) and cloned into a plasmid to form recombinant DNA before being packaged into adenoviruses, forming recombinant adenoviruses (rAAV).

- (a) (i)** Explain why PCR was used to obtain copies of the hTERTC27 instead of using mRNA for the reverse transcriptase.

[1]

- (ii)** Explain what information the scientists must have in order to use PCR to produce copies of hTERTC27.

[2]

- (b)** Describe how the recombinant DNA is formed.

[3]

Glioblastoma cells were injected subcutaneously (just below the skin) into three groups of mice. After a week, each group was treated differently as shown in the **Table 2.1**.

Group	Treatment
I	none
II	rAAV-EGFP - recombined AAV containing gene that codes for green fluorescent protein
III	rAAV-hTERTC27 - recombined AAV containing the sequence that codes for 251 amino acids from the C-terminal end of human reverse transcriptase

(Note: treatment was done by injecting directly into the tumor of the mice with rAAV.)

Table 2.1

The size of the tumor for each group was measured weekly and the volume calculated. The survival rates of these mice were also tracked. The 2 graphs below show the results obtained from the three groups of mice.

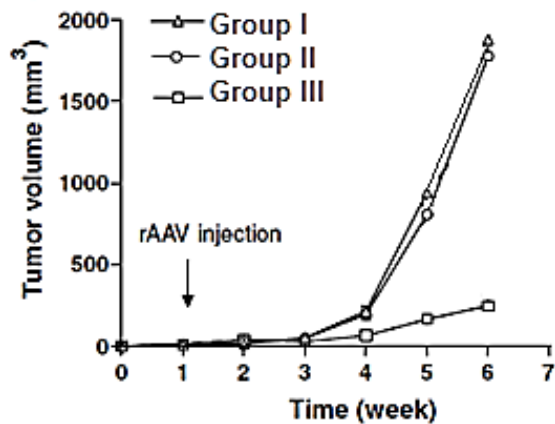


Fig. 2.2

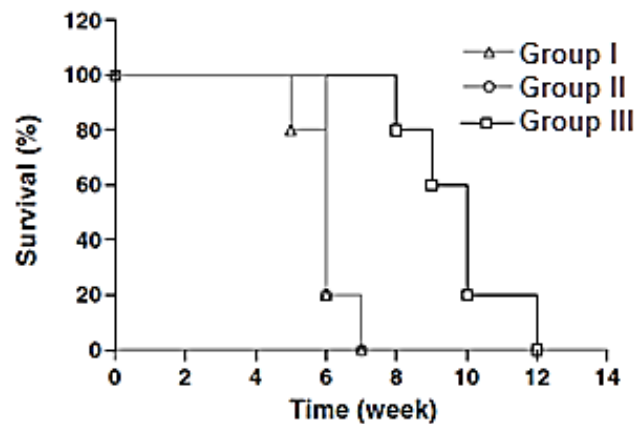


Fig. 2.3

(c) (i) What information can be obtained from the group II mice that have been injected with rAAV-EGFP?

(ii) Using information from both graphs, discuss the effectiveness of using gene therapy to control glioblastoma.

[3]

(d) In the course of their investigations, the researchers also found that the actions of hTERTC27 are only confined to the tumor cells and did not affect other cells in the body.
Suggest why this is an advantage.

[1]

[Total: 11]

- 3 'Golden Rice' is a transgenic plant introduced to help solve vitamin A deficiency in children. The daffodil gene encoding phytoene synthase (*psy*), one of the three genes used to develop 'Golden Rice', was hypothesized to be the limiting step in beta-carotene accumulation. Scientists systematically tested *psy* cDNAs from alternative plant sources by inserting each into maize cells. Maize was chosen because it was rich in the precursor that is required for beta-carotene synthesis.

Fig. 3.1a is a schematic diagram showing a part of the plasmid carrying the transgene used to transform the maize callus. The plate on the left in **Fig. 3.1b** shows the transformed maize calluses with the plasmid without *psy* gene (control). The plate on the right contains transformed maize calluses with the plasmid containing the popcorn *psy* gene. **Fig. 3.1c** is a histogram showing the carotenoid content of maize calluses transformed with a given *psy* gene (from daffodil (Np), popcorn (Zm), chilli pepper (Ca), or tomato (Le)).

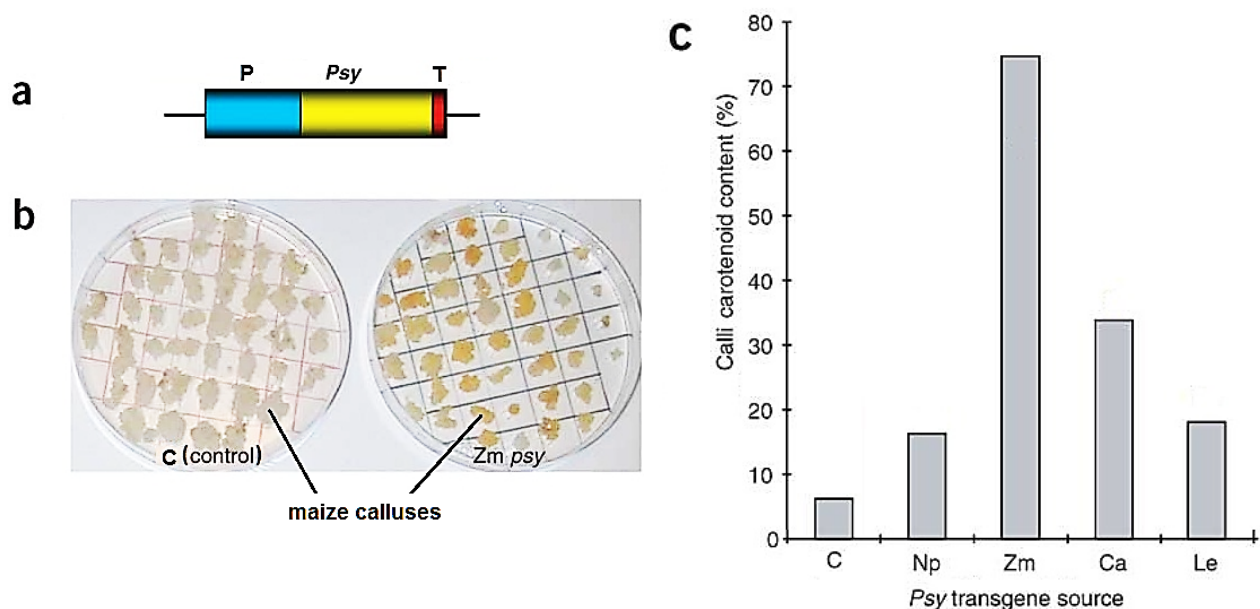


Fig. 3.1

- (a) (i) Explain what is meant by a 'transgenic plant'.

[1]

- (ii) Suggest why the daffodil gene encoding phytoene synthase is the limiting step in beta-carotene accumulation.

[2]

(b) With reference to **Fig. 3.1a**, explain the significance of having the following with the *psy* gene:

(i) a promoter (P)

[1]

(ii) a terminator (T)

[1]

(c) With reference to **Fig. 3.1b**, describe how a maize callus transformed with the plasmid containing the popcorn *psy* gene is formed.

[3]

(d) With reference to **Fig 3.1c**,

(i) evaluate the effectiveness of the various *psy* genes in increasing the carotenoid content of the maize calluses.

[2]

(ii) suggest an explanation for the differences in the carotenoid content of the maize calluses.

[2]

(e) Besides transgenic crop plants, man has also introduced transgenes into salmon to increase its yield. The biggest social concerns seem to focus on the impact that the transgenic fish could have on wild salmon populations should it escape.

(i) Describe the impact that transgenic fish have on wild salmon populations should it escape.

[2]

(ii) Suggest **one** way in which one can prevent the transgene from entering the gene pool of the wild salmon populations.

[1]

[Total: 15]

4 Planning question

Lately genetically modified (GM) foods are often in the news. While genetic modification have made improvements in many crops and helped to increase yields, many groups have raised loud protest against 'tinkering' with nature. Currently, GM foods before entering Singapore require certification by the Agri-food and Veterinary Authority (AVA) to ensure that they have undergone extensive testing for potential side-effects.

News reports surface recently a possible infiltration of GM corn into Singapore. This GM corn contains a gene coding for a toxin which is supposed to boost corn production by killing the pests feeding on it. Although the presence of the toxin was found to be harmless in humans thus far, it has created a public outcry.

To allay the fears of the general public, AVA conducts random testing on batches of corn from three of the most popular supermarkets in Singapore. As a key employee of AVA, you are tasked to investigate the presence of the gene coding for the toxin in different corn samples.

Your planning must be based on the assumption that you have been provided with the following equipment and materials:

Corn samples from three supermarkets

Centrifuge

Mortar and pestle

DNA extraction buffer

Polynucleotide synthesizer

DNA sequencing information of the gene coding for the toxin

Free deoxyribonucleoside triphosphates

Thermocycler

Microcentrifuge tubes

Agarose gel

Electrophoresis set-up

Loading dye

Ethidium bromide

UV lighting equipment

Micropipettes and sterile tips

Any normal laboratory glassware e.g. test-tubes, beakers, measuring cylinders, graduated pipettes, glass rods, etc.

Your plan should have a clear and helpful structure to include:

- An explanation of the theory to support your practical procedure
- A description of the method used to include the scientific reasoning behind the method
- The type of the data generated by the experiment
- How the results will be analysed including how the safety of consumption of corn can be determined

[Total: 12]

Free-response question

Write your answers to this question on the separate paper provided.

Your answer:

- should be illustrated by large, clearly labelled diagrams, where appropriate.
- must be in continuous prose, where appropriate.
- must be set out in section (a), (b), etc., as indicated in the question.

- 5 (a)** Describe how stem cells and gene therapy can be used to treat a named genetic disorder. Include in your answer the significance of using stem cells for the treatment. [9]
- (b)** Discuss the factors which prevent gene therapy from becoming an effective treatment. [6]
- (c)** Suggest to what extent genetic fingerprints are an example of discontinuous variation. [5]